

Hierarchical cEBMF for multi-omics

Model capacity, analytical benchmarks, and a defensible paper contribution

Research assessment for the cEBMF project

Literature checked: 14 September 2026

Assessment. This is a plausible basis for a methods paper if the contribution is a coherent, useful, and computationally effective solver for *learned conditional distributions between molecular programs*. Hierarchical multi-omics, empirical Bayes integration, nonlinear latent relationships, and interpretability each have substantial prior art. None, on its own, is a defensible novelty claim.

The most promising combination is:

Separate additive factorizations for each modality, connected by flexible conditional loading priors, with joint posterior inference that accounts for uncertainty in both upstream and downstream programs.

An EBNM-based computational construction, usable across the learned prior registry, could make this statistically and practically distinctive. It would need to work beyond the particular CGB-HMM toy example.

Relative to your existing work. EBMF already reduces factor updates to normal-means problems, and cEBMF already supplies flexible covariate-moderated priors [17, 18]. The additional contribution would be treating the informative covariates as uncertain latent programs inside a joint model, with the resulting feedback and parameter updates. Prior flexibility or EBNM modularity alone would not be new.

Closest comparisons. MOFA+ and JIVE are essential reference points, but a contemporary paper also needs EV-BIDIFAC for empirical Bayes linked matrix decomposition, CAVACHON for a hierarchical multi-omics generative model, and HALO for relationships between RNA and ATAC representations [1, 2, 7, 9, 10]. SIFA, MuVI, SOFA, and MEFISTO cover related uses of external information [11, 12, 13, 14].

What this document establishes. We compare model structure, derive exact results in controlled examples, identify the actual inference contribution, and specify an empirical benchmark. Numerical values in the analytical examples are population calculations, *not fitted-method performance*. No competing methods or new hierarchical solver were run for this report.

Scope. The proposed model below is the paired Gaussian observation model developed in the companion [joint-inference derivation](#). It is a proposal, not a statement that the current package implements full joint inference. The literature review is targeted rather than exhaustive; publication potential is a judgment, not a guarantee of novelty or acceptance.

1 The proposed model and the comparison target

For paired samples, let

$$Y^A \in \mathbb{R}^{N \times P_A}, \quad Y^R \in \mathbb{R}^{N \times P_R}.$$

Use separate loading matrices $A \in \mathbb{R}^{N \times K_A}$ and $R \in \mathbb{R}^{N \times K_R}$, with feature factors $F^m \in \mathbb{R}^{P_m \times K_m}$:

$$Y_{ip}^A \mid A, F^A \sim \mathcal{N} \left(\sum_{k=1}^{K_A} A_{ik} F_{pk}^A, (\tau_{ip}^A)^{-1} \right), \quad (1.1)$$

$$Y_{ip}^R \mid R, F^R \sim \mathcal{N} \left(\sum_{j=1}^{K_R} R_{ij} F_{pj}^R, (\tau_{ip}^R)^{-1} \right). \quad (1.2)$$

Only observed entries enter the likelihood. Samples are conditionally independent under the following row-wise prior:

$$p(A, R \mid \theta) = \prod_{i=1}^N \left\{ \prod_{k=1}^{K_A} g_k^A(A_{ik} \mid A_{i,<k}; \theta_k^A) \right\} \left\{ \prod_{j=1}^{K_R} g_j^R(R_{ij} \mid R_{i,<j}, A_i; \theta_j^R) \right\}. \quad (1.3)$$

Each g is normalized; a sparse subset of the displayed parents can be used. Fixed observed covariates may also be included. The feature-factor prior $p(F^A, F^R \mid \psi)$ is a separate design choice, including independent shrinkage or an HMM along a meaningful feature ordering.

Interpretation. The same ATAC state can change an RNA program’s activation probability, amplitude distribution, or uncertainty. Different RNA programs need not reuse identical ATAC scores. The model also permits dependencies among programs within each modality.

The generative ordering $A \rightarrow R$ does not restrict posterior inference to that direction. For fixed parameters,

$$p(A, R, F^A, F^R \mid Y^A, Y^R) \propto p(Y^A \mid A, F^A) p(Y^R \mid R, F^R) p(A, R) p(F^A, F^R).$$

RNA can therefore update ATAC. This property is shared by coherent joint latent-variable models in general; it is not unique to cEBMF.

Four different meanings of capacity.

1. *Reconstruction capacity*: which finite mean matrices can be fitted?
2. *Distributional capacity*: which joint laws and conditional relationships between programs can be represented?
3. *Statistical efficiency*: how much paired data is needed to learn those relationships and improve prediction?
4. *Inference capacity*: can the algorithm approximate the relevant posterior at useful scale and quantify uncertainty?

Greater distributional flexibility does not imply better finite-sample performance or easier inference.

2 Comparison with factorization methods

The following are structural comparisons of the cited formulations. An absent mechanism means it is not native to that formulation, not that the method could never be extended to include it.

Method	Native structure and strength	Relation to the proposal
JIVE (2013); AJIVE (2018)	Low-rank joint and individual sample subspaces; AJIVE estimates shared structure using angles and perturbation arguments.	A linear shared-subspace description, without a learned nonlinear conditional law between modality-specific scores in the original models [2, 3].
MOFA+ (2020)	Shared sample scores, view-specific feature weights, sparsity and ARD; factors can be shared, view-specific, or active in subsets. Supports groups, missing data and several observation likelihoods.	Already jointly borrows information and provides probabilistic inference. Its standard score-prior construction does not provide the proposed autoregressive conditional loading distributions [1].
SLIDE (2019); BASS (2016)	Partially shared components across subsets of views. BASS uses Bayesian structured shrinkage; SLIDE uses structured sparsity for the decomposition.	Shared/private structure and sparsity are existing capabilities, not distinctive benefits of a loading hierarchy [4, 5].
BIDIFAC+ (2022); EV-BIDIFAC (2024)	Low-rank modules supported on subsets of linked row and column groups. EV-BIDIFAC uses empirical variational Bayes shrinkage and supports imputation.	A direct baseline for an empirical Bayes integration claim. The proposed distinction is learning conditional program distributions, rather than only the sharing pattern and shrinkage of modules [6, 7].
ProJIVE (2026)	Probabilistic joint and individual factorization, estimated by maximum likelihood using EM.	A current probabilistic JIVE comparator; making JIVE probabilistic would not be a new contribution [8].

A fair reading of MOFA+ and JIVE

MOFA+ does *not* require every factor to affect every modality. JIVE explicitly includes individual variation. Therefore, allowing modality-specific factors is insufficient positioning.

For orientation, the shared-score Gaussian predictor is $Y^m = ZW_m^\top + E^m$. A shared/private linear formulation writes $Y^m = SU_m^\top + V_mB_m^\top + E^m$. These schematic equations describe matrix structure, not every prior, constraint, or likelihood in the methods above.

The useful new question is whether a nonlinear, uncertain relation between distinct programs can be learned more efficiently and interpreted more directly through Equation (1.3). It is not whether existing models can represent any view-specific signal at all.

3 Hierarchical and information-informed comparators

Method	Native structure and strength	Consequence for the paper
CAVACHON (2024 preprint; earlier 2022 workshop)	Hierarchical VAE based on a supplied modality DAG; Gaussian-mixture innovation latents feed parent-aware decoders.	Close prior art for biological hierarchy and conditional independence. A factor-level conditional-prior model must be distinguished from this decoder hierarchy [9].
HALO (2025)	RNA/ATAC representations linked through learned mappings and causal constraint losses, with an interpretable nonlinear additive decoder.	Neither hierarchical RNA–ATAC modeling nor interpretable nonlinear representations are new. Its published generative-prior and loss construction differs from Equation (1.3) [10].
SIFA (2017)	Joint and individual factors partly informed by observed covariates through nonparametric functions.	Covariate-informed nonlinear factor relationships already exist. Latent parents, their uncertainty, and feedback are the relevant distinction here [11].
MuVI (2023); SOFA (2026)	MuVI uses feature-set information through structured shrinkage. SOFA separates guided and unguided factors using sample-level guide variables.	Different ways to incorporate prior knowledge; include when the evaluation supplies pathways, annotations or observed drivers [12, 13].
MEFISTO (2022)	Gaussian-process structure on sample factors over continuous covariates such as time and space.	Structured factors are existing work. Its sample-axis covariance is different from an HMM over genomic features [14].
MultiVI (2023); GLUE (2022)	MultiVI handles paired and unpaired RNA/ATAC with a multimodal VAE. GLUE uses a feature guidance graph to link modality embeddings.	Relevant for single-cell integration, count data, missing modalities and unpaired settings. Their scope is broader than the present Gaussian paired-data proposal [15, 16].

Two distinctions that matter

CAVACHON’s generative equations put independent mixture innovation variables into a decoder that incorporates ancestors. In contrast, Equation (1.3) places dependencies in normalized distributions of scalar program loadings, while keeping feature reconstruction linear. This is a different parameterization and inductive bias, not proof of strictly greater model capacity.

HALO’s Gaussian latent priors and additional constraint/mapping losses (Methods, Equations 17–24) differ from a normalized $p(R | A)$. Neither comparison establishes superiority; both require experiments.

4 Analytical benchmark I: what hierarchy does not add

4.1 Any finite pair of fitted matrices has a shared-score representation

Let the fitted signals be

$$M^A = AF_A^\top, \quad M^R = RF_R^\top.$$

Define $Z = [A \ R]$ and view weights

$$W_A = [F_A \ 0], \quad W_R = [0 \ F_R].$$

Then

$$M^A = ZW_A^\top, \quad M^R = ZW_R^\top. \quad (4.1)$$

Thus a shared-score matrix parameterization with up to $K_A + K_R$ columns can reproduce the two fitted means exactly. The minimal joint rank can be smaller.

What follows. There is no universal in-sample reconstruction advantage merely from giving the modalities different loading matrices.

What does not follow. This construction does not match the probability law of Z , the method’s prior probability on the constructed weights, its identifiability constraints, or the efficiency of learning these weights from new data. It is not an equality of the full MOFA+ and hierarchical cEBMF model classes.

4.2 A linear Gaussian hierarchy reduces to shared/private structure

For a sample, use column-vector notation and suppose

$$a \sim \mathcal{N}(0, \Sigma_A), \quad r = Ba + \eta, \quad \eta \sim \mathcal{N}(0, \Sigma_\eta), \quad \eta \perp a.$$

Then

$$y^A = F_A a + \epsilon_A, \quad y^R = F_R Ba + F_R \eta + \epsilon_R.$$

The same a drives both views, and η is RNA-specific. Whitening a and η , when their covariance matrices are positive definite, yields independent standard Gaussian scores with transformed feature weights.

This is a shared/private linear-Gaussian factor model at the level of the observation distribution. Method-specific orthogonality and sparsity restrictions can still differ. Adding a linear Gaussian hierarchy alone therefore does not create the central expressive advantage being sought.

4.3 Flexible ordering is not a causal identification result

For unrestricted conditional families, any joint density can be factorized by the chain rule in any variable order. A dense DAG is then a parameterization, not evidence that ATAC causes RNA. Restricted families, sparse parents, biological anchors and the choice of factor orientation make the model informative, but introduce assumptions that must be assessed.

Factor scaling, signs, permutations, and sometimes rotations are non-identifiable without additional restrictions. Nonnegativity alone does not guarantee uniqueness. Graph edges between unanchored factors should not be presented as identified regulatory effects.

5 Analytical benchmark II: nonlinear dependence and shrinkage

5.1 A dependency invisible to affine cross-modal prediction

Let A_1, A_2 be independent Bernoulli(1/2) variables and define an RNA loading $R = A_1 \text{ xor } A_2$:

A_1	A_2	R	$\mathbb{E}R = \frac{1}{2}, \quad \text{Cov}(A_1, R) = \text{Cov}(A_2, R) = 0.$
0	0	0	
0	1	1	
1	0	1	
1	1	0	

The population least-squares predictor among affine functions $b_0 + b_1 A_1 + b_2 A_2$ is 1/2. Its mean squared error is $\text{Var}(R) = 1/4$. The nonlinear oracle

$$\mathbb{E}(R \mid A_1, A_2) = A_1 + A_2 - 2A_1 A_2$$

has zero error. This establishes a strict advantage over *affine prediction using these two observed inputs*.

A neural activation prior can approximate this gate. For example,

$$\pi_\kappa(a) = \text{sigmoid}\{\kappa(2|a_1 - a_2| - 1)\}, \quad R \mid a \sim (1 - \pi_\kappa(a))\delta_0 + \pi_\kappa(a)\mathcal{N}(1, d^2).$$

The Boolean model is approached as $\kappa \rightarrow \infty$ and $d^2 \rightarrow 0$; it is not identical to finite-variance CGB. A single linear-logistic gate cannot express XOR, whereas a suitable nonlinear network can.

Limits of the example. These numbers are not benchmark scores for fitted MOFA+, JIVE, or cEBMF. An expanded shared basis containing $A_1 A_2$ represents this mean relationship with a linear readout. Non-Gaussian latent models can also have nonlinear posterior predictors. No impossibility claim for every MOFA+ configuration follows.

5.2 Useful dependence can affect variance rather than mean

Suppose

$$R \mid A = a \sim \mathcal{N}(0, v(a)), \quad T = R + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, s^2)$$

with independent noise. Then $\mathbb{E}(R \mid A) = 0$, so knowing A cannot improve prediction of R from A alone under squared error. However,

$$\mathbb{E}(R \mid T, A = a) = \frac{v(a)}{v(a) + s^2} T, \quad \text{Var}(R \mid T, A = a) = \frac{v(a)s^2}{v(a) + s^2}.$$

ATAC can therefore improve RNA denoising by setting the amount of shrinkage, even without a conditional mean association.

This motivates conditional log scores and calibration alongside reconstruction error. A benchmark assessing only cross-modal mean prediction would miss an important use of conditional shrinkage.

6 Analytical benchmark III: can RNA improve ATAC?

6.1 An exact population result

Let U be an ATAC loading, a vector of loadings, or its noiseless signal, with finite second moment. Let D_A denote ATAC evidence and D_R additional RNA evidence. Under one fixed joint probability law, define

$$\hat{U}_A = \mathbb{E}(U \mid D_A), \quad \hat{U}_{AR} = \mathbb{E}(U \mid D_A, D_R).$$

The tower property and orthogonality of conditional expectation give

$$\boxed{\mathbb{E}\|U - \hat{U}_A\|^2 - \mathbb{E}\|U - \hat{U}_{AR}\|^2 = \mathbb{E}\|\hat{U}_{AR} - \hat{U}_A\|^2 \geq 0.} \quad (6.1)$$

To verify this, write $U - \hat{U}_A = (U - \hat{U}_{AR}) + (\hat{U}_{AR} - \hat{U}_A)$. The cross term has expectation zero after conditioning on both data sets.

Thus additional RNA evidence cannot increase the *integrated Bayes risk* of the optimal ATAC posterior mean. Improvement is strict exactly when the posterior mean changes with positive probability. Information can change higher moments without changing the mean. The analogous result holds with RNA and ATAC exchanged.

The result is not a guarantee for every realized sample, for an estimated empirical Bayes prior, or for a local variational solution. Nor does it compare two separately fitted, possibly misspecified models. It explains why bidirectional improvement is possible under a directed generative model.

6.2 Where the RNA evidence enters the actual update

For an ATAC loading $v = A_{ik}$, hold parameters fixed. Let Ω_i^A be the observed ATAC features for sample i . Its exact full conditional has the form

$$\begin{aligned} p(v \mid \text{rest}) &\propto \exp\{-\tfrac{1}{2}a_{ik}v^2 + b_{ik}v\} g_k^A(v \mid A_{i,<k}) \\ &\times \prod_{\substack{h>k \\ v \in \text{pa}(A_{ih})}} g_h^A(A_{ih} \mid A_{i,<h}) \prod_{\substack{j \\ v \in \text{pa}(R_{ij})}} g_j^R(R_{ij} \mid R_{i,<j}, A_i). \end{aligned} \quad (6.2)$$

Here $a_{ik} = \sum_{p \in \Omega_i^A} \tau_{ip}^A (F_{pk}^A)^2$ and

$$b_{ik} = \sum_{p \in \Omega_i^A} \tau_{ip}^A F_{pk}^A \left(Y_{ip}^A - \sum_{h \neq k} A_{ih} F_{ph}^A \right).$$

RNA observations enter indirectly through the RNA loadings and factors being inferred jointly. The final product in Equation (6.2) is the feedback missing when one only treats estimated ATAC loadings as fixed RNA covariates.

The full conditional of a directed model already conditions on children. Introducing an independently fitted reverse conditional does not in general produce the same joint model; the two conditionals must be compatible.

7 What an EBNM-based solver would contribute

7.1 A general interface, with real mathematical qualifications

For a scalar loading v , a factorized variational coordinate has

$$q^*(v) \propto \exp \left\{ -\frac{1}{2}av^2 + bv + \mathbb{E}_{q_{\text{pa}(v)}} \log g(v \mid \text{pa}(v)) + \sum_{c \in \text{ch}(v)} \mathbb{E}_{q_v} \log g_c(c \mid \text{pa}(c)) \right\}.$$

The coefficients a, b now come from the *expected* log likelihood: replace the realized squared factors by their second moments, and take expectations of the residual-factor products in b . All expectations hold v fixed. This is the standard variational coordinate identity, not a new general theorem. For spike-and-continuous priors, densities and updates use a common dominating measure with a zero atom.

For *exact full conditionals* with other latents realized, define the ordinary EBNM posterior at fixed parents and parameters,

$$q_0(v) \propto e^{-av^2/2+bv} g(v \mid \text{pa}(v)), \quad D(v) = \sum_{c \in \text{ch}(v)} \log g_c(c \mid \text{pa}(c)).$$

Then $p(v \mid \text{rest}) \propto q_0(v)e^{D(v)}$. An independent proposal $v' \sim q_0$ has Metropolis acceptance $\min\{1, \exp[D(v') - D(v)]\}$. With suitable support and ergodicity, this targets the required conditional; its finite-run accuracy depends on mixing. Refitting prior parameters inside this acceptance step would invalidate the argument. In variational inference, uncertainty in parents also requires the expected-own-prior correction detailed in the companion note.

7.2 The learned registry does not contain interchangeable model classes

Registry family	Conditional capacity relevant to the paper
cash, lcash, po_lcash	Mixture weights depend on covariates; centered scale mixtures change sparsity/variance but retain zero conditional prior mean.
cgb, cgb_sharp	Covariates change spike/slab probability, with global Gaussian slab parameters. Sharpening changes fitting, not a new generic hierarchy.
cgb_sharp_2	Separate positive-mean and negative-mean Gaussian slabs plus spike; slab means do not impose truncated support.
emdn, spiked_emdn	Covariate-dependent mixture weights, means and variances; the latter adds a zero atom. These are the broadest scalar families here.
hmm, hmm_pos, hmm_neg	Sequence priors, currently ignoring side covariates and using unit adjacency. They require block treatment; in this proposal they act on F .

For CGB with active probability $\pi(x)$, slab mean μ and variance d^2 , $\mathbb{E}(L \mid x) = \pi(x)\mu$ and $\text{Var}(L \mid x) = \pi(x)d^2 + \pi(x)(1 - \pi(x))\mu^2$. This is a gate, not unrestricted regression. The current registry's posterior moments alone are not enough for general child messages: density evaluation, sampling or integration, and coherent parameter learning are needed.

8 An empirical benchmark that would test the claims

These are proposed experiments, not predicted rankings. The central design is to vary *dependence type* independently of noise level, rank, sparsity, and feature smoothness.

Scenario	Question tested	Informative comparisons
Linear Gaussian shared/private factors	Does flexibility cost accuracy or runtime when unnecessary?	MOFA+, AJIVE/ProJIVE, EV-BIDIFAC; linear hierarchy.
Nonlinear activation: AND, XOR, smooth gates	Can conditional program structure improve denoising and prediction?	Conditional versus independent priors; HALO, CAVACHON; shared factors plus nonlinear regression.
Variance-only dependence	Does the method learn adaptive shrinkage and predictive dispersion?	Conditional scale mixture versus homoscedastic/independent priors.
Asymmetric SNR, reversed in a second experiment	Can downstream evidence help the noisy upstream modality?	Full joint versus one-way plug-in and no-child-message ablations.
No dependence; shuffled pairing; wrong graph	Can the method decline to borrow, and resist negative transfer?	Independent fits and sparsity-controlled hierarchy.
Three or more modalities with subset sharing	Does a DAG improve on established partial-sharing structure?	SLIDE, BASS, EV-BIDIFAC; matched total latent dimension.
Block features, smooth features, and permuted features	Is any gain due to the loading hierarchy or the HMM?	Same loading model with and without the feature prior; smoothing controls.
Realistic count noise, batch effects, missing modalities	What is the scope beyond normalized Gaussian matrices?	MultiVI; HALO/CAVACHON for paired hierarchy; explicit preprocessing and likelihood controls.

Required internal ablations

Use the same observation model and feature prior for: (i) independent cEBMF; (ii) one-way ATAC-to-RNA plug-in conditioning; (iii) one-way conditioning in the reverse direction; (iv) the proposed joint inference; (v) within-modality hierarchy only; (vi) a linear conditional prior. Also compare an independent factorization followed by a trained nonlinear map between loadings. That tests whether the benefit requires joint inference rather than just a flexible regression.

The current block-shaped toy data are valuable for debugging, but align closely with the HMM prior. They do not establish general multi-omics performance. Vary the feature prior independently of the loading model. RNA columns need a scientifically meaningful ordering before adjacency smoothing is interpreted biologically; unit index spacing is not genomic distance.

9 Evaluation, identifiability and computational evidence

Primary endpoints. In simulations, report noiseless-signal reconstruction error separately for ATAC and RNA, conditional prediction error in both directions, and uncertainty calibration. Assess program recovery after appropriate scale, sign and permutation alignment; use subspace recovery where individual factors are not identifiable. Log scores require predictive distributions; original JIVE does not natively provide them.

Three distinct holdouts.

1. Hold out entries for within-sample denoising.
2. Hold out an entire modality for test samples, inferring their loadings only from available modalities.
3. Hold out biological samples, donors or studies for transfer. For single-cell studies, avoid treating cells from the same donor as independent biological replications.

The corresponding estimands are different and should not be pooled. An entire-modality holdout on genuinely new samples is especially informative for learned conditional distributions.

Prevent leakage. Train transformations, feature selection, hyperparameters and conditional networks using training data; select settings with validation data. Test RNA must not enter the ATAC covariates used to predict that same held-out RNA. Parent uncertainty should be integrated, not replaced by information from the hidden modality. Distinguish transductive matrix completion from prediction for new samples.

Fairness. Tune ranks and regularization rather than fixing an arbitrary common K . A model with K_A ATAC factors and K_R RNA factors does not have the same complexity as a shared model with $\max(K_A, K_R)$ factors. Report both matched-capacity and independently tuned comparisons, network sizes, optimization budgets and restarts. Supply the same external information when comparing mechanisms that use it. When HMM smoothing is unique to one arm, report that advantage separately.

Comparable scales. Raw ELBO values from different model families are not a common performance metric. Compare held-out prediction for the same observed quantity and units. Gaussian losses on transformed expression are not directly comparable to count likelihoods. Use repeated simulation replicates with uncertainty intervals, and held-out donors/studies for real-data uncertainty. Clustering and enrichment can supplement, but cannot replace, prediction and calibration.

Computational evidence. A residual-based Gaussian sweep has a likelihood cost of order $N \sum_m P_m K_m$. The loading hierarchy adds child-prior evaluations and parameter optimization. With $K = K_A + K_R$, a dense ordered graph has $K(K-1)/2$ edges, so cost can grow rapidly with rank. Integration accuracy and network input size add further costs. An HMM with S states has the usual $O(P_m S^2)$ forward-backward cost per feature-factor sequence before exploiting transition structure. Report runtime, memory, convergence and, for Monte Carlo, effective sample size. No scaling advantage can be claimed from the update formula alone.

Suggested real-data scope. Start with two independent paired RNA-ATAC studies, preferably spanning different tissues or technologies. Add a three-modality application only if broad multi-omics capability is a central claim. For the first paper, a carefully delimited Gaussian analysis of processed molecular features is legitimate; raw-count performance needs an observation-model extension or a justified pipeline.

10 What would make this a convincing paper?

A defensible contribution statement

We develop a hierarchical empirical Bayes factorization that learns conditional distributions between modality-specific molecular programs. The model preserves additive feature contributions and supports joint posterior updates across a specified program graph. An EBNM-based inference strategy accommodates multiple conditional prior families. We evaluate when this structure improves estimation and when it causes negative transfer.

This is suggested positioning, not a claim that all of it has been implemented or validated.

The most substantial methodological contribution would be the solver and its statistical behavior. Standard full-conditionals, the variational coordinate identity, and the conditional-expectation risk identity are established tools. Their application is useful, but a paper needs more than writing them down.

A strong submission would provide:

1. A normalized joint model, clearly separating observed covariates, latent parents, feature structure, and learned hyperparameters.
2. An implemented common-objective inference algorithm, with mixed spike/continuous supports, uncertain-parent expectations and child feedback handled consistently. A small exact or accurately sampled reference should check the approximation.
3. Evidence across at least two materially different conditional prior families, for example CGB gates and mixture mean/variance models. Regularization and preprocessing must have a coherent role in the target.
4. A clear account of ordering, anchors, sparsity and factor identifiability; sensitivity to alternative orderings and incorrect graphs should be reported.
5. Fair comparisons against strong linear/EB methods and hierarchical neural comparators, with convincing positive examples and failures.

Useful theory targets. Characterize a restricted model class for which conditional program dependencies are identifiable, or prove approximation/error properties for the proposed EBNM message algorithm. An exact-coordinate ascent proof gives monotonicity only for updates that truly maximize one well-defined objective; it does not automatically cover Monte Carlo gradients, plug-in parents, sharpening heuristics, changing supports, or repeated black-box prior refits.

My judgment. The idea is promising as an interpretable statistical alternative for studying dependence between programs. Its clearest potential benefit over standard shared-factor models is a targeted conditional-distribution assumption, especially with nonlinear gates or covariate-dependent shrinkage. Its benefit over hierarchical VAEs would need to come from the particular additive model, prior flexibility, inference quality, data efficiency, or practical simplicity.

A paper showing only that one alternating two-fitter example improves RNA would be weak. A carefully validated general solver, a useful biological question, and evidence explaining *when and why* joint conditional modeling helps could constitute a substantial contribution.

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